

JNA to AKI transformation

V0.1

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1. User code Java

1.1. JNA

1.1.1 MainActivity.java

MainActivity.java in Android is transformed to index.ets and index.d.ts respectively as in section 1.2.1 index.ets and 1.2.2 index.d.ts.

```

public static double num = 100.00;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    binding = ActivityMainBinding.inflate(getLayoutInflater());
    setContentView(binding.getRoot());

    String name = "abcd";
    Memory strMemory = new Memory(name.length() + 1);
    strMemory.setString(0, name);
    change.C.changeName(strMemory);
    name = strMemory.getString(0);
    System.out.println("name1:" + name);

    String name2 = "abcd";
    name2 = change.C.changeName2();
    System.out.println("name2:" + name2);

    int age = 21;
    Memory ageMemory = new Memory(4);
    ageMemory.setInt(0, age);
    change.C.changeNum(ageMemory);
    age = ageMemory.getInt(0);
    System.out.println("after changed: " + age);

    num = change.C.changeDouble();
    System.out.println("num:" + num);

    System.out.println("final number: " + change.C.add1(5, 5));

    int area = MyRectangle.myRectangle.calculateArea(3, 4);
    int perimeter = MyRectangle.myRectangle.calculatePerimeter(3, 4);
    System.out.println("Area: " + area);
    System.out.println("Perimeter: " + perimeter);

```

```
AddNumber.MyLib.INSTANCE.addN(new AddNumber.MyLib.addNumber() {
    public int add(int num1, int num2) {
        return num1 + num2;
    }
});
```

```
AddNumber.MyLib.INSTANCE.show(new AddNumber.MyLib.addNumber1() {
    public String show(String s, int num) {
        return num + s;
    }
});
```

```
structExample.MyStructure.ByReference s = new structExample.MyStructure.ByReference();
s.a = 2;
s.b = 3;
int result = structExample.MyLibrary.lib.add(s);
System.out.println("Result: " + result);
```

```
}
```

```
}
```

```
interface change extends Library {
    change C = (change) Native.load("jnatoaki", change.class);
    void changeName(Pointer str);
    String changeName2();
    void changeNum(Pointer num);
    double changeDouble();
    int add1(int a, int b);
}
```

```
interface MyRectangle extends Library{
    MyRectangle myRectangle = (MyRectangle) Native.load("jnatoaki", MyRectangle.class);
    int calculateArea(int length, int width);
    int calculatePerimeter(int length, int width);
}
```

```

class structExample {
    public static class MyStructure extends Structure {
        public static class ByReference extends MyStructure implements Structure.ByReference {}
        public int a;
        public int b;
        @Override
        protected List<String> getFieldOrder() {
            return Arrays.asList("a", "b");
        }
    }
    public interface MyLibrary extends Library {
        MyLibrary lib = (MyLibrary) Native.load("jnatoaki", MyLibrary.class);
        int add(MyStructure.ByReference s);
    }
}

class AddNumber {
    public interface MyLib extends Library {
        MyLib INSTANCE = (MyLib) Native.load("jnatoaki", MyLib.class);

        interface addNumber extends Callback {
            int add(int num1, int num2);
        }
        interface addNumber1 extends Callback {
            String show(String s, int num);
        }
        void addN(addNumber callback);
        void show(addNumber1 callback);
    }
}

```

1.2 AKI

These index.ets and index.d.ts are transformed from section MainActivity.java.

1.2.1 index.ets

```

var num = 100.00;
num = testNapi.changeDouble();

testNapi.add1(5, 5)

function setName(n) {
    name = n;
}
var name = "abcd";
testNapi.changeName();
// @ts-ignore
testNapi.JSBind.bindFunction("set", setName)

// var name = new ArrayBuffer(64);
// var UintName = new BigUint64Array(name);
// name[0] = 'a';
// name[1] = 'b';
// name[2] = 'c';
// name[3] = 'd';
// testNapi.changeName(name);
// var test = UintName.toString();
var name2 = "abcd";
name2 = testNapi.changeName2();

var age = new ArrayBuffer(1);
var UintAge = new Uint8Array(age);
UintAge[0] = 21;
testNapi.changeNum(age)
var msg = UintAge.toString();
// var result = testNapi.changeNum(age);
// UintAge = new Uint8Array(result);
// var msg = UintAge.toString();

```

```
class AddNumber {
  add(num1: number, num2: number): number {
    return num1 + num2;
  }
  show(s: string, num: number): string {
    return num + s;
  }
}

// @ts-ignore
testNapi.JSBind.bindFunction("AddNumber.add", new AddNumber().add);
// @ts-ignore
testNapi.JSBind.bindFunction("AddNumber.show", new AddNumber().show);

var s = new testNapi.MyStructure();
s = testNapi.MakeMyStructure(2, 3);
var result = testNapi.add(s);
```



```

@Entry
@Component
struct Index {
  @State message: string = 'JNA'

  build() {
    Row() {
      Column() {
        Text(this.message)
          .fontSize(50)
          .fontWeight(FontWeight.Bold)
          .onClick(() => {
            hilog.info(0x0000, 'testTag', 'Test name = %{public}s', name);
            hilog.info(0x0000, 'testTag', 'Test name1 = %{public}s', name2);
            hilog.info(0x0000, 'testTag', 'Test num = %{public}d', num);
            hilog.info(0x0000, 'testTag', 'Test age = %{public}s', msg);
            hilog.info(0x0000, 'testTag', 'Test final number = %{public}d', testNapi.add1(5, 5));
            testNapi.addN();
            testNapi.show();
            var area = testNapi.calculateArea(3, 4);
            var perimeter = testNapi.calculatePerimeter(3, 4);
            hilog.info(0x0000, 'testTag', 'Test area = %{public}d', area);
            hilog.info(0x0000, 'testTag', 'Test perimeter = %{public}d', perimeter);
            hilog.info(0x0000, 'testTag', 'Test Result = %{public}d', result);
          })
      }
      .width('100%')
    }
    .height('100%')
  }
}

```

1.2.2 index.d.ts

Additionally, in TS, we need to define the *.d.ts file as well.

```
export const changeName2: () => string;
export const changeName: () => void;
export const changeNum: (num: ArrayBuffer) => void;
export const changeDouble: () => number;
export const add1: (a: number, b: number) => number;
export const addN: () => void;
export const show: () => void;
export const calculateArea: (length: number, width: number) => number;
export const calculatePerimeter: (length: number, width: number) => number;
export const add: (s: MyStructure) => number;
export class MyStructure {
  constructor();
}
export const MakeMyStructure: (a: number, b: number) => MyStructure;
```

2. User code CPP

2.1 JNA

2.1.1 native-lib.cpp

This native-lib.cpp is transformed to hello.cpp in section 2.2.1 hello.cpp.

```

#include <cstring>
#include <iostream>
#include <android/log.h>

#define TAG "Jacky"

#define LOGD(...) __android_log_print(ANDROID_LOG_DEBUG, TAG, __VA_ARGS__);

struct MyStructure {
    int a;
    int b;
};

extern "C" int add(MyStructure* s) {
    return s->a + s->b;
}

typedef char* (*Callback2)(char*, int);

extern "C" void changeName(char* str) {
    const char* newName = "aaa";
    strcpy(str, newName);
}

extern "C" char* changeName2() {
    char* newName = new char[4];
    strcpy(newName, "aaa");
    return newName;
}

extern "C" void changeNum(int* num) {
    (*num) += 10;
}

extern "C" double changeDouble() {
    return 60.00;
}

extern "C" int add1(int a, int b) {
    return a + b;
}

```

```

extern "C" int calculateArea(int length, int width) {
    return length * width;
}
extern "C" int calculatePerimeter(int length, int width) {
    return 2 * (length + width);
}

typedef int (*Callback)(int, int);
extern "C" void addN(Callback callback) {
    int i = callback(2, 3);
    LOGD("add number: %d", i);
}

extern "C" void show(Callback2 callback) {
    char* newName = new char[4];
    strcpy(newName, "pm");
    newName = callback(newName, 4);
    LOGD("current time: %s", newName);
}

```

2.2 AKI

2.2.1 hello.cpp

This hello.cpp is transformed from native-lib.cpp in section 2.1.1 native-lib.cpp.

```

#include <iostream>
#include <string>
#include <aki/jsbind.h>
#include <hilog/log.h>

struct MyStructure {
    int a;
    int b;
};

//    strcpy(str, newName);
//    str = newName;
//    aki::ArrayBuffer newName(str.GetData(), str.GetCount());
//    uint8_t* data = newName.GetData();
//    data[0] = 'a';
//    data[1] = 'a';
//    data[2] = 'a';
char* changeName2() {
    char* newName = new char[4];
    strcpy(newName, "aaa");
    return newName;
}

void changeNum(aki::ArrayBuffer num) {
    uint8_t *temp = num.GetData();
    temp[0] += 10;
}

//    aki::ArrayBuffer newNum(num.GetData(), num.GetCount());
//    uint8_t *temp = newNum.GetData();

int calculateArea(int length, int width) { return length * width; }
int calculatePerimeter(int length, int width) { return 2 * (length + width); }

void show() {
    auto f = aki::JSBind::GetJSFunction("AddNumber.show");
    auto newName = f->Invoke<char *>("pm", 4);
    OH_LOG_Print(LOG_APP, LOG_DEBUG, LOG_DOMAIN, "testTag", "Test current time: %{public}s", newName);
}

```

```

void addN() {
    auto f = aki::JSBind::GetJSFunction("AddNumber.add");
    auto i = f->Invoke<int>(2, 3);
    OH_LOG_Print(LOG_APP, LOG_DEBUG, LOG_DOMAIN, "testTag", "Test add number: %{public}d", i);
}

double changeDouble() { return 60.00; }

int add1(int a, int b) { return a + b; }

void changeName() {
    const char *newName = "aaa";
    auto f = aki::JSBind::GetJSFunction("set");
    f->Invoke<void>(newName);
}

MyStructure MakeMyStructure(int a, int b) {
    MyStructure myStructure = {a, b};
    return myStructure;
}

int add(MyStructure *s) { return s->a + s->b; }

JSBIND_GLOBAL() {
    JSBIND_FUNCTION(changeName);
    JSBIND_FUNCTION(changeName2);
    JSBIND_FUNCTION(changeNum);
    JSBIND_FUNCTION(changeDouble);
    JSBIND_FUNCTION(add1);
    JSBIND_FUNCTION(addN);
    JSBIND_FUNCTION(show);
    JSBIND_FUNCTION(calculateArea);
    JSBIND_FUNCTION(calculatePerimeter);
    JSBIND_FUNCTION(add);
    JSBIND_FUNCTION(MakeMyStructure);
}

JSBIND_ADDON(entry)

JSBIND_CLASS(MyStructure) {
    JSBIND_CONSTRUCTOR<>();
    JSBIND_PROPERTY(a);
    JSBIND_PROPERTY(b);
}

```

3. CMakeLists.txt file

7.1 JNA

```
# For more information about using CMake with Android Studio, read the
# documentation: https://d.android.com/studio/projects/add-native-code.html

# Sets the minimum version of CMake required to build the native library.

cmake_minimum_required(VERSION 3.22.1)

# Declares and names the project.

project("jnatoaki")

# Creates and names a library, sets it as either STATIC
# or SHARED, and provides the relative paths to its source code.
# You can define multiple libraries, and CMake builds them for you.
# Gradle automatically packages shared libraries with your APK.

add_library( # Sets the name of the library.
            jnatoaki

            # Sets the library as a shared library.
            SHARED

            # Provides a relative path to your source file(s).
            native-lib.cpp)

# Searches for a specified prebuilt library and stores the path as a
# variable. Because CMake includes system libraries in the search path by
# default, you only need to specify the name of the public NDK library
# you want to add. CMake verifies that the library exists before
# completing its build.

find_library( # Sets the name of the path variable.
             log-lib

             # Specifies the name of the NDK library that
             # you want CMake to locate.
             log)
```



```
# Specifies libraries CMake should link to your target library. You
# can link multiple libraries, such as libraries you define in this
# build script, prebuilt third-party libraries, or system libraries.
```

```
target_link_libraries( # Specifies the target library.
    jnatoaki

    # Links the target library to the log library
    # included in the NDK.
    ${log-lib})
```

7.2 AKI

```
# the minimum version of CMake.
cmake_minimum_required(VERSION 3.4.1)
project(AkiProject)

set(NATIVERENDER_ROOT_PATH ${CMAKE_CURRENT_SOURCE_DIR})

include_directories(${NATIVERENDER_ROOT_PATH}
    ${NATIVERENDER_ROOT_PATH}/include)

add_library(entry SHARED hello.cpp)

add_subdirectory(aki)
target_link_libraries(entry PUBLIC aki_jsbind)
```

4. Notes

In Android project we need to add JNA dependency like this in the build.gradle file.

```
dependencies {  
    implementation 'net.java.dev.jna:jna:5.6.0@aar'
```